1. Identity

Document	STAR-PKG-STAR-AUDIT-EN · v2.2 · 2026-11-02
Package	star-audit
Version	0.13.1
Capability	star.audit.geodesy
Maturity	Beta (VERIFIED)
Python	⩾3.9
Source commit	dc09f61
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_audit

2. Licence and obligations

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3. What it needs at run time

NEEDS	LICENCE	SOURCE
star-geodesy	Apache-2.0	S.T.A.R.
mpmath	BSD-3-Clause	https://mpmath.org/

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import
hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_audit-
0.13.1-py3-none-any.whl
```

FILE	SHA-256
star_audit-0.13.1-py3-none-any.whl	58b8cc1d42722232d5bec34affc6e3ce3187 bec737044a418fb6ab48f44259f2
star_audit-0.13.1-source.zip	1672cfee7bff35d84e5e711d23b8773330e4 5b09dc8cfbc8cb5fb601a45f6815

5. What it does, and its limits

Measures the accuracy envelope (per altitude band, 0-36,000 km) of any ECEF→geodetic library against an exact reference.

That these libraries are "wrong": they may be tuned for near-surface use; other call options were not audited.

- `audit_function(fn)` / `run_external(python, code)`: 450 points (10 latitudes x 5 longitudes x 9 altitudes from 0 to 36,000 km) with KNOWN geodetic truth; the library converts their exact ECEF; errors are measured against truth.
- Envelope per altitude band: max latitude error (deg), height error (m), 3-D position error (m); worst case.
- The library runs in its own interpreter (`run_external`); a failing library run raises, never returns a score.
- `audit(python, driver)`: drives a time library ONLY through its documented UTC-string → TAI-string API on 7 probes (incl. instants inside 23:59:60) and classifies OK / WRONG / REJECTED / PAST_TABLE_WARNED / PAST_TABLE_SILENT against a hand-written IERS Bulletin C truth.
- `audit(command, driver, cases)` vs Vallado SGP4-VER (33 cases, 667 epochs; fixtures with provenance): per-case max 3-D error, median, refusals. Cases are paired by ORDER (NORAD 20413 appears twice: short window and 3.5-year arc).
- `audit(impls, cases)`: every implementation runs in its own interpreter (driver `teme_to_gcrs(utc, r_km)`); probe validation first: Vallado's published TEME→GCRF example (2004-04-06 07:51:28.386009 UTC) must be reproduced to < 1 m; then pairwise divergence envelopes per year 1980-2050 on LEO and GEO vectors, and refusals.
- `audit(impls, cases)`: deterministic corpus (seed 20261004, sha256-pinned) of 40 valid TM frames (CCSDS 132.0-B, every header option: OCF, secondary header 1..8 octets) and 260

copies with 1..4 bit flips in header, data or FECF. Probe validation: an implementation counts only if it decodes every valid frame field-exact; a driver exception is never a credible rejection. Reports silent acceptance of bad-FECF frames, CRC collisions and acceptance disagreements.

- `audit(impls, epochs)`: each implementation in its own interpreter (`tdb_minus_tt(jd_tt)`); probe validation: every epoch within 50 us of the published low-precision formula (Explanatory Supplement, 0.001657 sin g + 0.000014 sin 2g); pairwise envelopes over 401 epochs 1900-2100, per century; refusals exclude.
- `audit(impls, epochs)`: each implementation in its own interpreter (`gmst_deg(jd_ut1)`); probe validation on Vallado's published Example 3-5 (GMST = 152.578787810 deg at 1992-08-20 12:14:00 UT1) within 0.1 arcsec; wrap-safe pairwise envelopes in arcsec over 201 epochs 1950-2050, per half-century; refusals exclude.
- `audit(impls, cases)`: rv → (p, e, i, RAAN, argp, nu) per implementation in its own interpreter; probe validation on Vallado's Example 2-5 (tolerances 0.01 km / 1e-5 / 0.01 deg, sized on the printed digits); seeded, sha256-pinned corpus of 60 regular + 20 singular (near-circular, near-equatorial, both) states; regular cases compared with the generating truth and pairwise; singular cases: each implementation's CONVENTION (0, sentinel, NaN, value, refusal).
- `audit(impls, cases)`: what TLE readers accept. Reference = the format rule (checksum mod 10 with '-' = 1, line numbers, equal satellite numbers, 69 columns). Corpus: the 29 well-formed SGP4-VER TLEs plus 6 derived kinds (field_digit, checksum, line_number, satnum_mismatch, truncated = must reject; resummed = well-formed, must accept), sha256-pinned. Probe validation: every valid TLE accepted, its inclination read within 1e-4 deg and its catalogue number read exactly (a NaN inclination or a wrong number is a misread: before 0.13.1 both were validated).
- `audit(impls, cases)`: what Space Packet decoders accept (CCSDS 133.0-B rule: version '000', length field = data octets - 1). Corpus (sha256 of the rounded corpus pinned): 40 valid + 10 idle (accept) and 4 × 40 defective (version, truncated, trailing, header_only: reject). Probe validation: every valid packet decoded field-exact.
- `audit(impls, cases)`: elliptic Kepler solvers M → E. Probe validation: Vallado Example 2-1 (M = 235.4 deg, e = 0.4 → E = 220.512074767522 deg, tolerance 1e-9 rad). Truth: 8 eccentricities (0 to 0.9999) x 41 anomalies, the exact inverse (40 digits, mpmath) of the double M sent to each library; circular error. A non-finite answer on a regular case is a refusal. Hostile inputs (NaN, inf, e = 1, e > 1, e < 0, huge or negative M) are recorded as a behaviour category per library. Corpus fingerprint pinned.
- `audit(impls, cases)`: zero-revolution Lambert solvers, driver `lambert(r1, r2, tof, mu) → v1` (prograde). Probe validation: Curtis Example 5.2 (v1 = (-5.9925, 1.9254, 3.2456) km/s, tolerance 1e-4 km/s). Truth without any Lambert solver: 315 cases built from known orbits (3 semi-major axes x 5 eccentricities up to 0.95 × 3 inclinations x 7 transfer angles from 10 to 330 deg); r1, r2 and the true v1 from the closed-form perifocal equations, time of flight from the forward chain nu → E → M. A non-finite or wrong-length answer is a refusal. Hostile inputs (NaN, tof <= 0, zero vector, 0 and 180 deg transfers) are recorded per library. Corpus fingerprint pinned.

- `audit(impls, cases)`: Kepler propagators, driver `propagate(r0, v0, tof, mu) → [x, y, z, vx, vy, vz]`. Probe validation: Vallado Example 2-4 (1e-3 km, 1e-5 km/s; the constant was cross-checked with NAIF prop2b). Truth without any propagator: 450 known orbits (3 semi-major axes x 5 eccentricities up to 0.95 × 3 inclinations x 5 transfer angles x 0 or 10 whole revolutions), states from the closed-form perifocal equations. A non-finite or wrong-length answer is a refusal. Each hostile input runs alone with a 120 s deadline; no answer is recorded as `error:Hang`.
- Every auditor reads the rows a driver printed through one guard: exactly one result per case, on the last output line, otherwise `RuntimeError`. Found by a hostile probe (2026-10-06): a reader that answered 1 case out of 203 was validated (the totals were counted on what came back), and an empty or non-JSON output raised undeclared exceptions.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_audit_0_13_1,
  author  = {Cavazzini, Andrea},
  title   = {{star-audit}: Geodetic conversion accuracy audit},
  version = {0.13.1},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_audit/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit dc09f61}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

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